



INDIAN INSTITUTE OF TECHNOLOGY MADRAS • BS DEGREE, DATA SCIENCE AND APPLICATIONS
BSMS4002 • DESIGN THINKING FOR DATA-DRIVEN APP DEVELOPMENT • PROJECT CLOSURE REPORT

PaisaPath

A financial mentor for first-time earners in India, one clear step at a time. This report covers the KDT-EAST phases we followed, the product that came out of them, and how the project was closed.

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Four phases, and what each one actually produced

Empathize

Nine interviews, nine journey maps. We talked to young Indians at three stages of handling money: a pre-earner on an allowance (Riya, 19), a cautious first earner (Samik, 23) and an early investor (Anuj, 23) among them. We mapped each person's journey from income to decision. All nine had tried to track their spending and given up. All nine raised trust or privacy before we asked. None could explain how to divide a salary. Every map bottomed out at the same point: **"I'll decide later."**

Analyze

Multi-why analysis on those symptoms landed on four root causes. A guidance gap (no beginner help at the moment income starts), a structure gap (our nine users needed four different budgeting approaches), a confidence gap (no way to see the reasoning behind a suggestion, so the decision gets put off) and a trust gap (unclear data practices and incentives). We framed these as three how-might-we questions: make first-money decisions clear and explainable, track and budget with minimal effort, and earn trust without replacing professional advice.

Solve

Fifteen ideas, each required to name the root cause it closes. Two were dropped on purpose: a SEBI-adviser marketplace (it needs real verified professionals to mean anything) and a peer tips community (unmoderated advice would undercut the trust position everything else depends on). One rule was fixed before any code was written: **the model never produces a number.**

Test

Tested as a working app, not a mockup. Ten screens run on seeded demo data. Two review rounds shaped the design. The sharpest question we got was "won't your AI become one more conflicting advisor?", and the answer is in the architecture: fixed rules compute every figure, and the model only writes prose about them. Assert-based self-checks pin the money maths, the six-profile classifier and the provider layer, and a screenshot pipeline walks every screen on each build.

Fifteen features, three tests they have to pass

PLAN & TRACK

- **Unified dashboard:** five accounts, bills, goals and net worth on one page
- **Automatic categorisation:** rules first, AI only for what rules cannot place
- **Budget generator:** a life-stage template bent toward your 3-month averages
- **Goal & emergency fund planner:** every goal carries its monthly number
- **Monthly review:** the month written as prose, not a report

DECIDE

- **Safe-to-Spend:** the hero number, its arithmetic one click away
- **What-if simulator:** a plain-words question becomes a 24-month projection
- **Context-aware nudges:** nine triggers, each naming the evidence that fired it
- **Mentor chat:** streamed answers that already know your figures
- **Health score:** five pillars, labelled self-reported until 20 real transactions

TRUST & LEARN

- **Onboarding profiler:** 22 questions, six profiles, each shown with its reasons
- **Draggable salary split:** always sums to exactly the salary
- **"Why am I seeing this?":** why, benefit, risk and two alternatives, on everything
- **Trust Centre:** permissions, disclosures and every AI call ever made
- **Lessons in context:** any dashed underline explains itself where you are

Human desirability EVIDENCE-LED

Every feature traces to something a named interviewee said. All nine journey maps stalled at the same task, comparing options, and that is where the simulator sits. The why-drawer answers the trust concern all nine raised unprompted. The app answers the question first earners actually have: what should I do next, and why should I believe you.

Business viability HONEST READ

Unmonetised by choice: no affiliate links, sponsored placements or referral fees, and the app says so. Cost to serve is low: the maths is local deterministic code and the AI layer accepts whichever engine is cheapest, including free local models. A flat subscription, or distribution through employers and colleges, fits the trust position; commissions on financial products would contradict it.

Technological feasibility SETTLED BY BUILDING IT

One runtime dependency (Express); React and Vite to build; SQLite from the Node standard library. The finance engine imports nothing, so it is fully testable. Twelve AI providers sit behind one interface with a deterministic fallback, so every screen still works when no AI is reachable.

What we used, what we finished, what we left

TOOLS USED

- **Method:** persona and journey-map canvases, multi-why analysis, how-might-we framing, open brainstorming, two structured review rounds
- **Build:** React with Vite, Express, SQLite via node:sqlite, hand-written SVG charts, assert-based Node self-tests
- **AI engines:** agent CLIs (claude, codex, gemini), local model servers, and hosted APIs. Twelve providers over three wire formats behind one interface
- **Evidence:** a Chrome DevTools Protocol pipeline that screenshots every screen and writes the PDF walkthroughs

ACCOMPLISHMENTS

- 10 screens built and running, onboarding through Trust Centre
- 22 questions score three axes into six financial profiles, each pinned by a worked-example test
- 9 nudge triggers, each dismissible and carrying its evidence
- 0 numbers written by AI: a test asserts the shown ledger sums to the shown figure
- 12 interchangeable AI engines, every call audit-logged, with a rules fallback when none answer

CLOSURE REMARKS

- **Descoped, with reasons on record:** the adviser marketplace and the peer community; both cut against the trust argument rather than for it
- **Left for a future team:** taxes, insurance choice, credit score tools
- **What held up under review:** separating "we measured this" from "you told us this", and keeping every number out of the model
- **Closed with:** working software, passing self-checks, seeded demo data and a documented codebase

From "I'll decide later" to one clear step, with the reason shown.